



10683 - Heavy elements following the giant flare from the magnetar SGR 1806-20

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		NIRSpec IFU Spectroscopy	(2) SGR1806-20

ABSTRACT

Giant flares from highly magnetic neutron stars (magnetars) have recently come to prominence as a possible site for the synthesis of the heaviest elements via the rapid capture of neutrons -- the r-process. This process is responsible for producing many elements heavier than iron (e.g., gold, iodine, thorium), but the astrophysical sites where it occurs have remained shrouded in mystery. Evidence for r-process production in magnetars comes from a strong MeV component following the giant flare from SGR 1806-20 in 2004, which is best modelled as the radioactive decay of newly synthesised heavy elements. Here, we propose to test heavy element production following the flare, making use of NIRSPEC IFU spectroscopy of the environs of SGR 1806-20 to directly search for heavy element signatures either in a flare remnant or fallback disc around the magnetar. These observations can provide direct evidence of a new site for r-process nucleosynthesis, while also providing one of the first optical/IR spectra of a magnetar, a goal of substantial scientific interest in its own right.

OBSERVING DESCRIPTION

We will obtain a NIRSPEC IFU observation of SGR 1806-20. We will use the IFU with the 4 point nod pattern to map a $\sim 5 \times 5$ arcsecond region around the magnetar. The observations will provide a high signal-to-noise ratio (>10 across most of the spectral window) of the magnetar counterpart itself, while also providing a sensitive search for any heavy elements produced during the merger. There are a modest set of orient constraints included in order to avoid diffraction spikes from nearby stars crossing the magnetar position.

Proposal 10683 - Targets - Heavy elements following the giant flare from the magnetar SGR 1806-20

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(2)	SGR1806-20	RA: 18 08 39.3370 (272.1639042d) Dec: -20 24 39.85 (-20.41107d) Equinox: J2000		
	Comments: Category=Star Description=[Neutron stars]				
Fixed Targets	(3)	SGR_refstar	RA: 18 08 38.5093 (272.1604554d) Dec: -20 24 40.14 (-20.41115d) Equinox: J2000		
	Comments: Category=Star Description=[Neutron stars]				

Proposal 10683 - Observation 1 - Heavy elements following the giant flare from the magentar SGR 1806-20

Observation	Proposal 10683, Observation 1										Tue May 19 18:00:17 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRSspec IFU Spectroscopy											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(2)	SGR1806-20	RA: 18 08 39.3370 (272.1639042d) Dec: -20 24 39.85 (-20.41107d) Equinox: J2000									
	Comments: Category=Star Description=[Neutron stars]											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	3 SGR_refstar	WATA	SUB2048	F110W	NRSRAPID	3	1	1	3.628	305021	
Template	HFF Readout Mode											
	false											
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-NOD										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G235M/F170LP	NRSIRS2RAPID	45	2	false	true	NONE	4	8	5368.712	

Proposal 10683 - Observation 1 - Heavy elements following the giant flare from the magnetar SGR 1806-20

Special Requirements	Aperture PA Range 228.97164917 to 248.97164917 Degrees (V3 90.0 to 110.0)
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